Pergamon

*Computers chem. Engng* Vol. 20, Suppl., pp. S1209–S1214, 1996
Copyright © 1996 Elsevier Science Ltd
Printed in Great Britain. All rights reserved
S0098-1354(96)00209-8
0098-1354/96 $15.00+0.00

# FLEXIBILITY, RELIABILITY AND MAINTENANCE CONSIDERATIONS IN BATCH PLANT DESIGN UNDER UNCERTAINTY

E. N. PISTIKOPOULOS[1] , T.V. THOMAIDIS, A. MELIN and M. G. IERAPETRITOU

Centre for Process Systems Engineering
Department of Chemical Engineering
Imperial College, London SW7 2BY, U.K.

**Abstract** - The paper addresses the problem of including aspects of uncertainty in process parameters, product demands and equipment availability at the design stage of multiproduct/multipurpose batch plants. A two-stage stochastic programming formulation is proposed with an objective function comprising installing, operating and maintenance cost, expected revenues from product sales and a penalty term accounting for expected losses due to unfilled orders. It is shown that the exploitation of the special structure of the mathematical model enables the reformulation of the problem to a large scale optimization program, featuring explicit scheduling constraints, in which parameter variability and equipment failures are simultaneously considered. A decomposition-based solution procedure is proposed to determine the optimal design, schedule and maintenance plan maximizing an expected profit criterion.

## INTRODUCTION

While issues related to the operating pattern (scheduling) of batch plants have been extensively studied in the open literature, the problem of accounting for equipment (un)availability and systems effectiveness under changing conditions has received much less attention despite its enormous practical significance (Straub and Grossmann, 1992; Subrahmanyam et al., 1994; Sanmarti et al., 1995). Recently, Ierapetritou and Pistikopoulos (1995a,b) considered the problem of batch plant design and operations with uncertain demands; however, equipment deterioration and maintenance considerations were not taken into account. Thomaidis and Pistikopoulos on the other hand in their FRAMS development (1994, 1995a,b), have proposed generic integrated tools for the simultaneous assessment of operability objectives, such as flexibility, reliability and availability. The aim of this work is to integrate such operability objectives in the design and scheduling of multiproduct/multipurpose batch plants.

The paper is organized as follows. First, we show how reliability aspects can be formally introduced in stochastic batch plant design and scheduling formulations; availability considerations are also included based on Markovian corrective maintenance principles. Finally, we briefly discuss how condition-based maintenance can be implemented based on an operability criticality analysis. A multiproduct batch plant design model is used throughout the text to illustrate the concepts (for ease in the presentation).

## BATCH PLANT DESIGN AND FLEXIBILITY-RELIABILITY OPTIMIZATION

Consider a simplified multiproduct batch plant design model involving N products, operating in single campaign without intermediate storage, in M stages (such as the one shown in Figure 1 involving two products and three stages). Equipment are assumed to be available at discrete sizes; typical reliability models are associated with each equipment to describe its deterioration as a function of time (for example, a Weibull function); forecasted product demands are described by given probability distributional forms.

The consideration of feasibility and equipment deterioration in the batch plant design stage can be achieved based on the following ideas. First, for feasibility consideration, a numerical integration scheme, such as a Gaussian Quadrature formula, is employed to approximate the multiple integral of the expected

[1]To whom correspondence must be addressed; email: e.pistikopoulos@ic.ac.uk

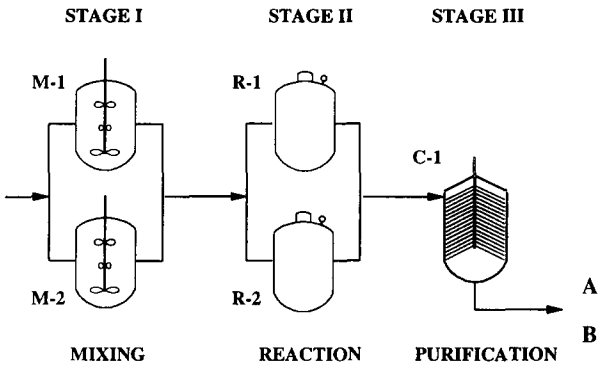



Figure 1: Batch Plant

profit term (comprising revenues and investment cost) in the space of uncertain demands. Furthermore, demand constraints are relaxed to allow for the feasibility property of Ierapetritou and Pistikopoulos (1995b) to hold, thereby enabling the integration of expected profit performed within the entire range of uncertainty; an implication of this is that the integration grid points can be defined "a priori". For reliability considerations, all feasible operating states are identified and then determined. Moreover, time horizon is divided into a number of time periods t, t=1,..,$N_T$, to account for equipment deterioration as a function of time. The objective function is also augmented to account for all time periods and all operable system states. These considerations lead to the following mathematical model; denoted as problem (P).

$$\textbf{(P)}\ \max_{y_{jsn}} \sum_{k\in K_{OP}} PT_k \sum_q w^q \sum_i p_i Q_i^q \quad \text{Expected Revenue}$$
$$-\delta \sum_j \sum_s \sum_n n a_j v_{js}^{\beta_j} y_{jsn} \quad \text{Investment Cost}$$
$$-\gamma \sum_{k\in K_{OP}} PT_k \sum_q w^q \Big(\sum_i p_i D_i^q - \sum_i p_i Q_i^q\Big) \quad \text{Penalty term for unfilled orders}$$

$$s.t.\ n_i^q \geq \sum_s \sum_n \frac{Q_i^q S_{ij}}{v_{js}} y_{jsn}\ ;\ \forall i,j,q \quad \text{Mass balances}$$
$$T_i^q \geq \sum_s \sum_n \Big(\frac{t_{ij}}{n}\Big) w_{ijsn}^q\ ;\ \forall i,j,q \quad \text{Cycle time constraints}$$
$$n_i^q = \sum_s \sum_n w_{ijsn}^q\ ;\ \forall i,j,q \quad \text{Logical constraints}$$
$$w_{ijsn}^q \leq U_{ijsn} y_{jsn}\ ;\ \forall i,j,s,n,q \quad \text{Logical constraints}$$
$$\sum_i T_i^q \leq H\ ;\ \forall q \quad \text{Horizon constraints}$$
$$Q_i^q \leq \theta_i^q\ ;\ \forall i,q \quad \text{Demand constraints}$$
$$PT_k = \sum_t^{N_T} w^t P_k(t) \quad \text{Average state probability}$$
$$P_k(t) = \pi_k \prod_j r_j(t)^{n_j^k} q_j(t)^{m_j - n_j^k} \quad \text{State probability over time}$$
$$\pi_k = \prod_j \frac{(m_j)!}{(n_j^k)!(m_j - n_j^k)!} \quad \text{Aggregator factor for state k}$$

In (P) $n_i^q$ is the number of batches of product i; H the time horizon; $S_{ij}$ the size factors (volume of a vessel in stage j required to produce one mass unit of product i); $t_{ij}$ the processing time of product i at stage j; $Q_i^q$ the production of product i; $\theta_i^q$ the uncertain demand of product i following a distribution $J(\theta_i)$; $p_i$ the price of product i; $\alpha_j$, $\beta_j$ are the cost coefficients of batch equipment at stage j; $\gamma$ penalty coefficient used to control the effect of penalty term; $\delta$ the coefficient used to annualize the capital cost; $v_{js}$ the available sizes of equipment at stage j; $n$ the number of equipment per stage; $y_{jsn}$ 0-1 variable denoting the selection of n equipments of size $v_{js}$ at stage j; $r_j(t), q_j(t)$ are the reliability and unavailability function of equipment at stage j; $m_j$ is the maximum number of operable equipment per stage j; $n_j^k$ is the number of operable equipment per stage j; $K_{OP}$ is the number of operable states.

Problem (P) is an MILP optimization model, the solution of which leads to the determination of an optimal design-schedule that maximize an expected profit taking into account uncertainty product demands

as well as equipment deterioration over time. Note that more involved scheduling constraints can be directly included in problem (P). However, since no maintenance tasks are considered, it is expected that (P) will generally result in conservative designs. In the next section corrective maintenance policies will be included in (P) to obtain less conservative designs.

## BATCH PLANT DESIGN AND MAINTENANCE OPTIMIZATION

The incorporation of corrective maintenance, following Markovian theory principles, requires the estimation of state probabilities through the solution of frequency balance equations (Billinton and Allan, 1983). However, at the design stage and since a corrective maintenance plan is not fixed but to be determined, parametric expressions of the frequency balance equations are needed. Such expressions can be derived based on the maintenance superstructure model of Thomaidis and Pistikopoulos (1994), which is shown in Figure 2. This then transforms problem (P) into the following problem (P1).

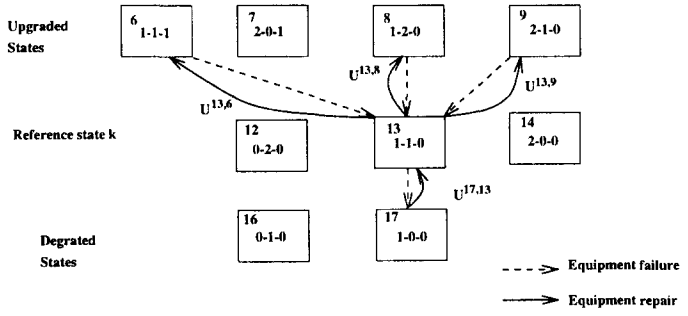



Figure 2: Maintenance superstructure for the system of Figure 1

$$\textbf{(P1)} \max_{y_{jsn}} \sum_{k\in K_{OP}} \pi_k P_k \sum_q w^q \sum_i p_i Q_i^q - \delta \sum_j \sum_s \sum_n n a_j v_{js}^{\beta_j} y_{jsn}$$
$$- \sum_{k\in K} \pi_k \{HP_k [C_{Lk} + C_{Ok}] + f_k [C_{Ck} + C_{Rk}]\} \quad \text{Maintenance cost}$$

subject to:

$$\left.\begin{aligned}
& n_i^q \geq \sum_s \sum_n \frac{Q_i^q S_{ij}}{v_{js}} y_{jsn} \; ; \; \forall i,j,q \\
& T_i^q \geq \sum_s \sum_n (\frac{t_{ij}}{n}) w_{ijsn}^q \; ; \; \forall i,j,q \\
& n_i^q = \sum_s \sum_n w_{ijsn}^q \; ; \; \forall i,j,q \\
& w_{ijsn}^q \leq U_{ijsn} y_{jsn} \; ; \; \forall i,j,s,n,q \\
& \sum_i T_i^q \leq H \; ; \; \forall q \\
& Q_i^q \leq \theta_i^q \; ; \; \forall i,q
\end{aligned}\right\} \text{Process model}$$

$$\left.\begin{aligned}
& \underbrace{\sum_{m\in M^k} Z^{k,m} \sum_{j\in J^k} (n_j^m - n_j^k)\mu_j + \sum_{l\in L^k} P_k \sum_{j\in J^l} (n_j^k - n_j^l)\lambda_j}_{\text{Rate Out}} = \\
& \underbrace{\sum_{m\in M^k} P^m \sum_{j\in J^k} (n_j^m - n_j^k)\lambda_j + \sum_{l\in L^k} Z^{l,k} \sum_{j\in J^l} (n_j^k - n_j^l)\mu_j}_{\text{Rate In}} \\
& \pi_k = \prod_{j\in J^k} \frac{(m_j)!}{(n_j^k)!(m_j - n_j^k)!}
\end{aligned}\right\} \text{Frequency balance equations}$$

$$\left.\sum_{k\in K} \pi_k P_k = 1\right\} \text{Normalising equation equations}$$

$$\left.\begin{aligned}
& \sum_{m\in M^k} U^{k,m} \sum_{j\in J^k} (n_j^m - n_j^k) nc_j \leq NC \quad k \in K \\
& f_k = \sum_{m\in M^k} Z^{k,m} \sum_{j\in J^k} (n_j^m - n_j^k)\mu_j + \sum_{l\in L^k} P_k \sum_{j\in J^l} (n_j^k - n_j^l)\lambda_j
\end{aligned}\right\} \text{Maintenance resources constraints}$$

$$\left.\begin{array}{l} Z^{k,m} - U^{k,m} \leq 0 \\ -Z^{k,m} + P_k - (1 - U^{k,m}) \leq 0 \\ Z^{k,m} - P_k \leq 0 \end{array}\right\} \text{ Glover Constraints}$$

$U^{k,m}$ 0-1 variables denoting the repair task from state k to state m; $Z^{k,m}$ artificial variables due to mathematical reformulations (Glover constraints); $NC$ maximum number of maintenance recourses; $nc_j$ required maintenance resources for maintenance of equipment j; $\lambda_j, \mu_j$ failure and repair rate of equipment j, respectively; $f_k$ frequency of state k. Note that a maintenance cost terms added to the objective function; $C_{Lk}, C_{Ok}$ denote labour and overhead cost associated with state k, respectively; $C_{Ck}, C_{Rk}$ denote call cost and cost of replaced equipment at state k, respectively.

Problem (P1) corresponds to a MINLP formulation due to the presence of bilinear terms in the objective function. However, a novel and efficient algorithmic procedure has been developed for its solution based on the following points:

(a) due to the feasibility property (Ierapetritou and Pistikopoulos 1995), the estimation of revenue is independent of the state in which the system currently resides (operable or degraded)- this significantly reduces the model size

(b) for fixed maintenance policy ($U^{k,m}$), the frequency balance equations in (P1) can be solved independently to determine the state probabilities ($P_k$)

(c) for fixed number of equipment per stage, the determination of operable states does not depend on batch plant design - this implies that design optimization and maintenance optimization can be performed independently (the algorithm requires one iteration to converge, see also example problem).

Based on these properties, the following iterative procedure is proposed for the solution of problem (P1).

Algorithmic Procedure

**Step 1:** Select an initial design $y_{jsn}$ and production $Q_i$. Set K=1, the lower bound $EP_L = -\infty$, the upper bound $EP_U = +\infty$ and select a tolerance $\epsilon$.

**Step 2:** Solve problem (P1) to obtain the optimal maintenance policy and the expected profit EP. Update the lower bound $EP_L^K = \max\{EP, EP_L\}$.

**Step 3:** Solve problem (P1) for the obtained maintenance policy to determine the optimal design and an upper bound $EP_U = \min\{EP, EP_U\}$.

**Step 4:** Check for convergence, if $EP_U \leq EP_L^K + \epsilon$; $y_{jsn}^K$ is the optimal design; go to Step 2 to determine the optimal corrective maintenance policy and stop. Otherwise, set K=K+1; return to Step 2 and continue until convergence.

The algorithm is guaranteed to converge in a finite number of iterations, however, no claim could be made regarding global optimality. Note also that Step 3 provides an upper bound to the solution of (P1) since it corresponds to the dual representation of the original problem. Extensions towards the inclusion of condition based preventive maintenance can be envisaged based on the criticality analysis step, which is described in the next section.

## CRITICALITY ANALYSIS

The objective of the criticality analysis is to identify the relative importance of equipment (un)availability over a time horizon by taking into account process interactions, reliability models for equipment deterioration, parameter variations and operational characteristics of the plant. In the following, the feasibility-reliability criticality index recently introduced by Thomaidis and Pistikopoulos (1995), is used as such a measure. It is based on the feasibility-reliability index (FR(t)) which represents the average operability of the design taking into account continuous uncertainty and equipment availability, by considering all the potential operating system states that occur as equipments fail. Each system state has an associated probability $P_k$ of occurrence which is a function of reliability model and reliability data of equipment (Billinton and Allan, 1983). In mathematical terms this index can be expressed as follows (Thomaidis and Pistikopoulos, 1995):

$$FR(t) = \sum_{k \in K} P_k(t) SF_k, \quad P_k(t) = \pi_k \prod_j r_j(t)^{n_j^k} q_j(t)^{m_j - n_j^k}, \quad SF_k = \int_{\theta \in T_k} J(\theta) d\theta$$

where $r_j(t), q_j(t)$ are the reliability and unavailability function of equipment at stage j; $m_j$ is the maximum number of operable equipment per stage j; $n_j^k$ is the number of operable equipment per stage j; $K_{OP}$ is the number of operable states; $SF_k$ is the stochastic flexibility index, proposed by Pistikopoulos and Mazzuchi (1990). Since we are interested in the sensitivity of the system to the occurrence of an event $l^*$, the feasibility-reliability-criticality index can be obtained as a derivative of $FR(t)$ with respect to the probability of the event $\rho_{l*}$ as follows:

$$FRC_{l^*}(t) = \sum_{k \in K^{OP}} \left\{ \frac{\partial P_k(t)}{\partial r_{l^*}} \right\}$$

$$\frac{\partial P_k(t)}{\partial r_{l^*}(t)} = \frac{\partial P_k(t)}{\partial \rho_m} \frac{\partial \rho_m}{\partial r_{l^*}}$$

$$\frac{\partial P_k}{\partial \rho_m} = \begin{cases} \prod_{j \in \{\overline{J}_k \setminus m\}} \rho_j \prod_{j \in J_k} (1-\rho_j), & \text{if } m \in \overline{J}_k \\ -\prod_{j \in \overline{J}_k} \rho_j \prod_{j \in \{J_k \setminus m\}} (1-\rho_j), & \text{if } m \in J_k \end{cases}$$

$$\frac{\partial \rho_m}{\partial r_{l^*}} = \prod_{c \in \{MC_m \setminus MC_m^*\}} \left[ 1 - \prod_{l \in c} r_l \right] \sum_{c^* \in MC_m^*} \left\{ \prod_{l \in \{c^* \, l^*\}} r_l \prod_{c \in \{MC_m^* \setminus c^*\}} \left[ 1 - \prod_{l \in c} r_l \right] \right\}$$

where $r$ denotes the probability of event; $\rho$ the probability of failure; $J$ and $\overline{J}$ are the sets of operable and failed equipments, respectively; subscripts $m$, $l$ denote the subsystem in aggregated system and a basic event respectively whereas superscript $*$ denotes the event that is considered for criticality analysis.

The estimation of the $FRC_{l^*}(t)$ for each piece of equipment $r_l$ at each time period could then be used to guide preventive/condition-based maintenance activities.

## ILLUSTRATING EXAMPLE

Consider the batch plant design of Figure 1 involving two products to be processed in three stages with two units per stage. Size factors, processing times, cost data as well as reliability data for equipment and maintenance cost are given in Table 1. The time horizon is 8000 hr. All required equipments are available in discrete sizes from the set $v_{js} \in \{3000, 3150, 3200, 3300, 3500\}$. The demand of both products are considered as uncertain parameters described by normal distribution functions of the form N(200,10) and N(100,10) for products A and B, respectively. Five quadrature points are used for each uncertain parameter for the evaluation of expected revenue.

Table 1: Example Data

(a) Size factors

| | Stage | | |
|---|---|---|---|
| Product | 1 | 2 | 3 |
| A | 2000 | 3000 | 4000 |
| B | 4000 | 6000 | 3000 |

(b) Processing times

| | Stage | | |
|---|---|---|---|
| Product | 1 | 2 | 3 |
| A | 8 | 20 | 8 |
| B | 16 | 4 | 4 |

(d) Prices of Products

| Product | $p_i$ |
|---|---|
| A | 5.5 |
| B | 7.0 |

(c) Investment cost coefficients

| Stage | $a_j$ | $b_j$ |
|---|---|---|
| 1 | 5 | 0.6 |
| 2 | 5 | 0.6 |
| 3 | 5 | 0.6 |

(d) Weibull parameters $weif\{\frac{u}{\alpha};\beta\}$

| $\alpha_j$ | $\beta_j$ |
|---|---|
| 65000 | 0.5 |
| 80000 | 0.6 |
| 110000 | 0.8 |

(e) Maintenance cost

| |
|---|
| 4 |
| 2.8 |
| 5.2 |

Using GAMS/CPLEX for batch plant design optimization (without any penalty term $\gamma = 0$), requires 1.17 CPU sec in SPARC 10 to obtain the optimal design (3000, 3000, 3000) having an expected profit of 1104.06 units, investment cost 914.8 units and expected revenue 2018.8 units.

When equipment deterioration is also considered (assuming failure rates described by $weibull(\frac{t}{\alpha};\beta)$, see Table 1) the solution of problem (P) using GAMS/CPLEX requires 4.53 CPU sec in SPARC 10 to find the optimal design (3200, 3200, 3200) with an expected profit of 789.9 units and investment cost of 950.9 units which corresponds to a more conservative estimation of the optimal design. This is mainly because the system reliability reduces as time increases due to lack of any repair mechanism. The estimation of system flexibility-reliability as a function of time is shown in Figure 3, with an average value over time of 0.871 which means that the system can withstand 87.1% of both demand variations and equipment failures. The incorporation of potential corrective maintenance policies results in the determination of an optimal

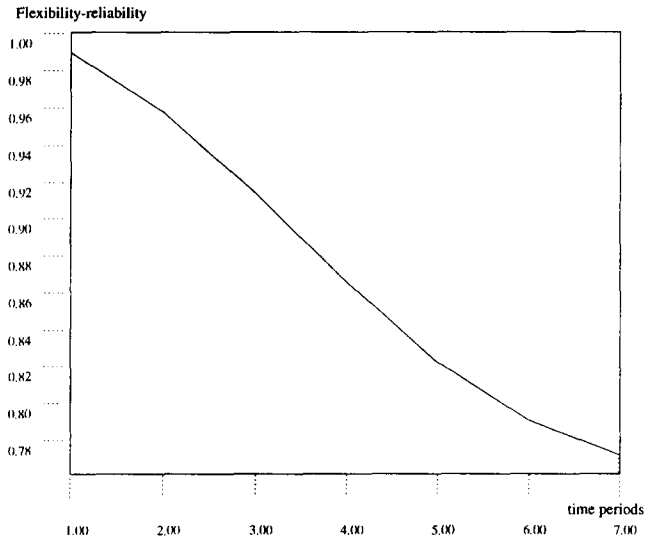



Figure 3: System Flexibility-Reliability

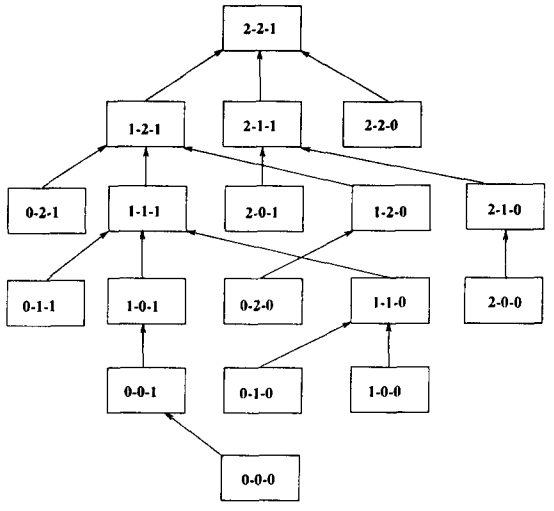



Figure 4: Optimal Maintenance Plan

less conservative design (3150,3150,3150) as well as the optimal maintenance plan as shown graphically in Figure 4. Finally, the results of the criticality analysis (shown in Table 2) suggest the purificator as the most critical equipment over time (with the mixers coming next and reactors last).

Table 2: Results of criticality analysis

| | t=0 | | | t=4000 | | |
|---|---|---|---|---|---|---|
| Equipment | FRC | Scaled FRC | Order of importance | FRC | Scaled FRC | Order of importance |
| mixer1 | -0.00012 | 0.000124 | purif | -0.1999 | 0.215110 | purif |
| mixer2 | -0.00012 | 0.000124 | mixer2 | -0.1999 | 0.215110 | mixer2 |
| reactor1 | -0.00002 | 0.000018 | mixer1 | -0.1354 | 0.145713 | mixer1 |
| reactor2 | -0.00002 | 0.000018 | reactor2 | -0.1354 | 0.145713 | reactor2 |
| purif | -1.00000 | 1.000000 | reactor1 | -0.9295 | 1.000000 | reactor1 |

## REFERENCES

Billinton, R. and Allan R.N., 1983, Reliability Evaluation of Engineering Systems: Concepts and Techniques. Prenum Press, New York and London.

Ierapetritou, M. G. and Pistikopoulos, E.N., 1995a, Design of Multiproduct Batch Plants with Uncertain Demands. *Comp. Chem. Engng.* **19S** S627-S632.

Ierapetritou, M. G. and Pistikopoulos, E.N., 1995b, Batch Plant Design and Operations under Uncertainty. Accepted for publication, *Ind. Chem. Engng. Res.*

Pistikopoulos, E.N., 1995, Uncertainty in Process Design and Operations *Comp. Chem. Engng.* **19S** S553-S563.

Pistikopoulos, E. N. and Mazzuchi T.A., 1990, A Novel Flexibility analysis Approach for Processes with Stochastic Parameters. *Comp. Chem. Engng.*, **14**, 991.

Sanmarti E., Espuna A. and Puigjaner L., 1995, Effects of Equipment Failure Uncertainty in Batch Production Scheduling. *Comp. Chem. Engng.* **19S** S565-S570.

Straub, D.A. and Grossmann I.E., 1992, Evaluation and Optimization of Stochastic Flexibility in Multiproduct batch Plants. *Comp. Chem. Engng.* **16** 69.

Subrahmanyam, S., Pekny J.F. and Reklaitis G.V., 1994. Design of Batch Chemical Plants under Market Uncertainty. *Ind. Eng. Chem. Res.* **33** 2688.

Thomaidis, T. V. and Pistikopoulos E.N., 1995a, Optimal Design for Flexible and Reliable Process Systems. IEEE Trans. Rel. **44**(2) 243-250.

Thomaidis, T. V. and Pistikopoulos E.N., 1995b, Towards the Incorporation of Flexibility, Maintenance and Safety in Process Design. *Comp. Chem. Engng.* **19S** S687-S692.

Thomaidis, T. V. and Pistikopoulos E.N., 1994, Flexibility, Reliability and Maintenance in Process Design. Foundation of Computer Aided Design, *Proceedings of FOCAPD, Snowmass, AIChE Symposium Series*, 304, **91**, 260-265.